

Answer: C

4 $F_{\text{net}} = F_{\text{propelling}} - F_{\text{friction}} = ma$

$F_{\text{propelling}}$ is constant

F_{friction} is not constant and is decreasing due to mass of tanker decreasing

m = mass of tanker is not constant and is decreasing with time

Hence F_{net} is not constant and is increasing

Therefore, a increases non-uniformly

Answer: A

5 For elastic collision, total kinetic energy is conserved